

DEEP INTO DEVOPS

The Ultimate Master Interview Guide

300+ REAL-TIME SCENARIOS AND QUESTIONS

Covering Cloud, Infrastructure, Automation, and Reliability

AWS & Azure Kubernetes Terraform
CICD Pipelines DevSecOps SRE Principles

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2026 Edition

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1. General & Cultural Interview Questions

1. Tell me about yourself and your journey into DevOps.
2. What is your daily routine as a DevOps Engineer?
3. How do you stay updated with the latest DevOps trends?
4. Describe a time you had a conflict with a Developer. How was it resolved?
5. What do you do when a production deployment fails at 2 AM?
6. How do you prioritize tasks when everything is marked as "Urgent"?
7. Explain DevOps to a non-technical person.
8. What is the "Three Ways" of DevOps (Gene Kim)?
9. Why did you choose the AWS/Cloud path?
10. What are your expectations from this role?
11. How do you handle a situation where a manager asks for a "quick fix" that bypasses security?
12. What is the most complex architecture you have worked on?
13. How do you manage work-life balance in a 24/7 on-call environment?
14. What is your favorite DevOps tool and why?
15. How do you mentor junior engineers in your team?

2. AWS Architecture & Global Infrastructure

1. Explain the difference between Regions and Availability Zones.
2. What is the significance of the "Edge Location"?
3. How do you design a highly available architecture on AWS?
4. What is a VPC (Virtual Private Cloud)?
5. Difference between Public Subnet and Private Subnet.
6. What is an Internet Gateway (IGW)?
7. Explain the role of a NAT Gateway.
8. Difference between Security Groups and NACLs.
9. What is VPC Peering? What are its limitations?
10. Explain AWS Transit Gateway.
11. How do you connect an On-Premise data center to AWS? (Direct Connect vs VPN).
12. What is Route 53? Explain its routing policies (Weighted, Failover, Latency).
13. How do you handle DNS for a global application?
14. What is the purpose of an Elastic IP?
15. Explain the concept of "Infrastructure as Code" in the context of AWS CloudFormation.
16. How do you secure data in transit within a VPC?
17. What is the default limit of VPCs per region?
18. Explain VPC Endpoints (Interface vs Gateway).
19. How do you share resources across AWS Accounts? (RAM - Resource Access Manager).
20. What is an Application Load Balancer (ALB) vs Network Load Balancer (NLB)?

AWS Storage Deep Dive

21. What are the S3 storage classes? (Standard, IA, Glacier, Deep Archive).
22. Explain S3 Bucket Policies vs IAM Policies.
23. What is S3 Object Lock?
24. How does S3 Cross-Region Replication work?
25. What is EBS (Elastic Block Store)?
26. Difference between GP2, GP3, IO1, and IO2 volumes.
27. What is EFS (Elastic File System) and when to use it over EBS?
28. How do you backup an EBS volume? (Snapshots).
29. Explain AWS Snowball and Snowmobile.
30. What is AWS Storage Gateway?
31. How do you move 100TB of data from On-Premise to S3 efficiently?
32. Explain S3 Versioning and its benefits.
33. What is a pre-signed URL in S3?
34. How do you enable MFA Delete in S3?
35. Explain AWS Backup service.

3. Advanced AWS Services & Troubleshooting

1. What is AWS Lambda? How is it billed?
2. Explain the "Cold Start" problem in Lambda.
3. What are Lambda Layers?
4. How do you trigger a Lambda function from an S3 upload?
5. Explain AWS Step Functions.
6. What is DynamoDB? Is it SQL or NoSQL?
7. Explain Partition Key and Sort Key in DynamoDB.
8. What is Amazon RDS? Mention supported engines.
9. How do you perform a Zero-Downtime database migration to AWS?
10. What is Amazon Aurora? How is it different from standard RDS?
11. Explain Multi-AZ vs Read Replicas in RDS.
12. What is AWS Elastic Beanstalk?
13. Explain AWS ECS (Elastic Container Service) vs EKS (Elastic Kubernetes Service).
14. What is Fargate?
15. How do you monitor AWS account activity? (CloudTrail).
16. Difference between CloudWatch Metrics and CloudWatch Logs.
17. What is Amazon GuardDuty?
18. How to protect against DDoS on AWS? (AWS Shield/WAF).
19. Explain AWS Organizations and SCPs (Service Control Policies).
20. What is AWS Systems Manager (SSM) and how do you use it to manage EC2?

AWS Real-Time Troubleshooting

21. You cannot SSH into your EC2 instance. What are the 5 things you check?
22. An EC2 instance is showing 100% CPU. How do you investigate?
23. Your ELB is returning 504 Gateway Timeout. What does this mean?
24. Users are complaining of slow S3 downloads. How to optimize? (CloudFront).
25. CloudFormation stack is stuck in "DELETE_FAILED". What do you do?
26. Auto Scaling is not launching new instances despite high load. Why?
27. A Lambda function is timing out. How to fix?
28. You lost your EC2 Key Pair. How do you regain access?
29. Billing is showing high costs for "Data Transfer Out". How to reduce?
30. RDS instance is in "Storage Full" state. How to recover?

4. Linux Internals & Networking

1. What is the Linux Kernel?
2. Explain the Boot process: BIOS -> MBR -> GRUB -> Kernel -> Init.
3. What is the difference between /etc and /var directories?
4. How do you check the running processes in Linux? (top/htop/ps).
5. What is a "Zombie Process"? How to kill it?
6. Explain Load Average (1, 5, 15 minutes).
7. How to check disk I/O performance? (iostat).
8. What is a "File Descriptor" in Linux?
9. Explain the difference between 'su' and 'sudo'.
10. How do you change file ownership and permissions? (chown/chmod).
11. What are the sticky bits and SUID?
12. How to view the last 100 lines of a log file continuously? (tail -f).
13. Explain 'grep', 'sed', and 'awk' with one-liner examples.
14. How do you find files modified in the last 24 hours?
15. What is the difference between a shell and a terminal?
16. Explain the 'PATH' environment variable.
17. How to check open ports on a server? (netstat/ss).
18. What is the purpose of /etc/hosts file?
19. How to troubleshoot a "Connection Refused" error?
20. Explain the 'dig' and 'nslookup' commands.

Networking for DevOps

21. Explain the OSI Model (7 Layers).
22. Difference between TCP and UDP.
23. How does a 3-way handshake work in TCP?
24. What is HTTP vs HTTPS?
25. Explain the DNS resolution process (Recursive vs Iterative).
26. What is a Subnet Mask?
27. Explain CIDR notation (e.g., /24).
28. What is a Reverse Proxy? (Nginx/HAProxy).
29. How to capture network packets? (tcpdump/wireshark).
30. What is a Firewall? (iptables/firewalld).
31. Explain Load Balancing algorithms (Round Robin, Least Connection).
32. What is a SSL/TLS Handshake?
33. How to check the route taken by a packet? (traceroute).
34. Explain MTU (Maximum Transmission Unit).
35. What is Port Forwarding?

5. Git, Branching & Collaborative Development

1. What is Distributed Version Control?
2. Explain 'git clone' vs 'git fork'.
3. What is the difference between 'git pull' and 'git fetch'?
4. Explain 'git merge' vs 'git rebase'. When to use which?
5. What is a 'merge conflict'? How to resolve it?
6. Explain the 'git stash' command.
7. What is 'git cherry-pick'?
8. How do you undo the last commit? (git reset/git revert).
9. Difference between 'git reset --soft', '--mixed', and '--hard'.
10. What is 'git reflog'?
11. Explain 'git bisect'.
12. How do you ignore files in Git? (.gitignore).
13. What is a Git Hook? Give an example (pre-commit).
14. Explain GitFlow branching strategy.
15. What are 'tags' in Git? (Lightweight vs Annotated).
16. How to squash multiple commits into one?
17. What is 'HEAD' in Git?
18. Difference between Git and GitHub/GitLab.
19. How to check who modified a specific line in a file? (git blame).
20. What is a Pull Request (PR) and the code review process?

6. Jenkins & The CI/CD Lifecycle

1. What is Continuous Integration (CI)?
2. What is Continuous Delivery (CD)?
3. How do you install Jenkins?
4. Explain the Jenkins Master-Slave architecture.
5. What is a Jenkinsfile? (Declarative vs Scripted).
6. Explain 'Stages', 'Steps', and 'Post' in a pipeline.
7. How do you trigger a Jenkins job? (Webhook, Cron, Poll SCM).
8. What are Jenkins Plugins? Name 5 essential ones.
9. How to secure Jenkins? (RBAC).
10. What are 'Parameters' in Jenkins?
11. How do you manage secrets/credentials in Jenkins?
12. Explain 'Global Tool Configuration'.
13. What is a Multi-branch Pipeline?
14. How do you backup Jenkins configurations?
15. What is 'Blue Ocean' in Jenkins?
16. How to integrate Jenkins with Docker?
17. What is a Shared Library in Jenkins?
18. How to handle parallel execution in a pipeline?
19. Explain the 'Agent' directive in a pipeline.
20. How to notify Slack/Email after a build failure?

Jenkins & CICD Scenarios

21. Your build is stuck in the queue. How do you investigate?
22. How to handle a pipeline that requires manual approval before Prod?
23. How to run a Jenkins job inside a Docker container?
24. What is 'Garbage Collection' in the context of Jenkins builds?
25. How to trigger a downstream job only if the upstream job succeeds?
26. Explain the concept of 'Artifact Repository' (Nexus/Artifactory).
27. How do you perform a rollback in your current project?
28. What is 'Self-healing' in a CI/CD pipeline?
29. How to track which developer broke the build?
30. Explain 'Canary Release' through Jenkins.

7. Docker: Containerization Deep Dive

1. What is a Container? How is it different from a VM?
2. Explain the Docker Architecture (Client, Host, Registry).
3. What is a Docker Image?
4. Explain the layers in a Docker Image.
5. What is a Dockerfile?
6. Explain the difference between 'CMD' and 'ENTRYPOINT'.
7. Difference between 'ADD' and 'COPY'.
8. How do you reduce the size of a Docker image? (Multi-stage builds).
9. What is '.dockerignore'?
10. Explain Docker Networking (Bridge, Host, None).
11. What are Docker Volumes? (Anonymous, Named, Bind Mounts).
12. How to see the logs of a container? (docker logs -f).
13. Explain 'docker-compose'.
14. What is the difference between 'docker stop' and 'docker kill'?
15. How to cleanup unused Docker objects? (docker system prune).
16. What is Docker Hub?
17. Explain 'docker inspect'.
18. How to run a container in the background? (-d).
19. What is a "Dangling Image"?
20. How to limit CPU/Memory for a container?

Docker Security & Advanced

21. How do you scan Docker images for vulnerabilities? (Trivy/Snyk).
22. What is 'Docker Content Trust'?
23. Can we run multiple processes in a single container?
24. What is 'Distroless' image?
25. Explain 'Docker Swarm' vs 'Kubernetes'.
26. How do you persist data in Docker?
27. What is 'Docker Init'?
28. Explain the concept of 'Container Orchestration'.
29. How to troubleshoot a "CrashLoopBackOff" in a container (even outside K8s)?
30. How to push an image to AWS ECR?

8. Kubernetes: Orchestration at Scale

1. What is Kubernetes (K8s)?
2. Explain K8s Architecture: API Server, Etcd, Scheduler, Controller Manager.
3. What is a 'Pod'? Can a Pod have multiple containers?
4. What is a 'ReplicaSet'?
5. Explain 'Deployment' vs 'StatefulSet'.
6. What is a 'DaemonSet'?
7. Explain 'Services' in K8s (ClusterIP, NodePort, LoadBalancer).
8. What is an 'Ingress' and 'Ingress Controller'?
9. Explain 'ConfigMap' and 'Secret'.
10. What are 'Liveness', 'Readiness', and 'Startup' probes?
11. Explain 'Namespaces'.
12. What are 'Labels' and 'Selectors'?
13. Explain 'Persistent Volume (PV)' and 'Persistent Volume Claim (PVC)'.
14. What is a 'StorageClass'?
15. Explain 'HPA' (Horizontal Pod Autoscaler).
16. What is 'VPA' (Vertical Pod Autoscaler)?
17. Explain 'Taints' and 'Tolerations'.
18. What is 'Node Affinity'?
19. Explain the 'Kubelet' and 'Kube-proxy'.
20. What is 'Kubectl'? Name 5 common commands.

Kubernetes Troubleshooting & Helm

21. How do you troubleshoot a Pod in 'Pending' state?
22. What to check when a Service is not reachable?
23. How to perform a 'Rolling Update' and 'Rollback' in K8s?
24. What is 'Helm'? Why is it called the K8s Package Manager?
25. Explain 'Helm Charts', 'Templates', and 'Values'.
26. What is 'RBAC' (Role-Based Access Control)?
27. Explain 'Network Policies'.
28. What is 'Sidecar Container' pattern?
29. Explain 'Init Containers'.
30. How to drain a node for maintenance?
31. What is 'Etcd' backup and restore?
32. Explain 'Admission Controllers'.
33. What is 'CRD' (Custom Resource Definition)?
34. Explain 'Service Mesh' (Istio/Linkerd).
35. How do you monitor a K8s cluster? (Prometheus/Grafana).

9. Infrastructure as Code (Terraform)

1. What is Infrastructure as Code (IaC)?
2. Why Terraform over CloudFormation or Ansible?
3. Explain the Terraform Workflow: Init -> Plan -> Apply -> Destroy.
4. What is the 'Terraform State' file? Why is it important?
5. What is 'State Locking'? How to achieve it? (DynamoDB).
6. Explain 'Variables' (input) and 'Outputs' in Terraform.
7. What are 'Data Sources'?
8. Explain 'Terraform Modules'.
9. What is a 'Provider' in Terraform?
10. Difference between 'count' and 'for_each'.
11. What is 'Terraform Workspace'?
12. Explain 'terraform.tfvars' file.
13. What is a 'Backend' in Terraform? (S3/Azure Blob).
14. How do you manage secrets in Terraform?
15. Explain 'terraform import'.
16. What is 'terraform refresh'?
17. How to target a specific resource for an update? (-target).
18. What is 'terraform taint'?
19. Explain 'Local-exec' vs 'Remote-exec' provisioners.
20. What is 'Terraform Cloud'?

Terraform Real-Time Scenarios

21. Your state file is corrupted. What do you do?
22. How to handle drift in Terraform?
23. You need to move a resource from one module to another. How? (state mv).
24. How to handle multi-region deployments in a single Terraform config?
25. What are 'Meta-arguments' in Terraform?
26. How to validate your Terraform code? (validate/fmt).
27. Explain 'Terraform Lifecycle' hooks (prevent_destroy, ignore_changes).
28. How to create 50 S3 buckets with different names using Terraform?
29. How to integrate Terraform with Jenkins?
30. What is 'Checkov' or 'Terrascan'?

10. Ansible: Configuration Management

1. What is Ansible? Is it Agent-based or Agentless?
2. Explain Ansible Architecture.
3. What is an 'Inventory' file? (Static vs Dynamic).
4. Explain 'Playbooks', 'Plays', and 'Tasks'.
5. What is a 'Module' in Ansible?
6. Explain 'Idempotency'.
7. What are 'Roles' in Ansible?
8. Explain 'Ansible Galaxy'.
9. What is 'Ansible Vault'?
10. Difference between 'Ad-hoc commands' and 'Playbooks'.
11. How do you handle variables in Ansible?
12. Explain 'Handlers'.
13. What is 'Ansible Facts'?
14. How to use 'Templates' (Jinja2) in Ansible?
15. Explain 'Ansible Tower' / 'AWX'.
16. How to run a playbook against a specific host?
17. What is the 'Become' keyword?
18. How to handle errors in playbooks? (ignore_errors, failed_when).
19. Explain 'Tags' in Ansible.
20. How to optimize Ansible performance for large inventories?

Ansible Real-Time Scenarios

21. You need to patch 500 Linux servers. How do you do it with Ansible?
22. How to check if a configuration change will work without actually applying it? (`-check`).
23. How to pass sensitive passwords to a playbook securely?
24. What is a 'Dry Run' in Ansible?
25. How to use Ansible to provision AWS resources?
26. Explain 'Conditionals' (when) in Ansible.
27. How to loop through a list of items? (`with_items/loop`).
28. What is 'Ansible Lint'?
29. How to debug a playbook execution? (`-vvv`).
30. Can Ansible replace Terraform? Discuss.

11. Monitoring, Logging & Observability

1. What is the difference between Monitoring and Observability?
2. Explain the 4 Golden Signals (Latency, Traffic, Errors, Saturation).
3. What is Prometheus? How does it collect data? (Pull vs Push).
4. Explain 'PromQL'.
5. What is 'Alertmanager'?
6. What is Grafana? How to create a dashboard?
7. Explain ELK Stack (Elasticsearch, Logstash, Kibana).
8. What is 'Filebeat' and 'Metricbeat'?
9. Explain Centralized Logging architecture.
10. What is 'Sentry' or 'New Relic'?
11. How do you monitor Kubernetes? (kube-state-metrics).
12. What is 'Tracing'? (Jaeger/Zipkin).
13. Explain 'Log Rotation'.
14. How to monitor application-specific metrics? (Micrometer/Prometheus client).
15. What is an 'Exporter' in Prometheus? (Node Exporter).

Monitoring & Logging Scenarios

17. Your Kibana dashboard is empty. Where do you check?
18. Prometheus is consuming too much memory. How to optimize?
19. How to set up an alert for "Disk Space > 90%" in Grafana?
20. How to track a single request across multiple microservices?
21. Explain 'Blackbox Monitoring' vs 'Whitebox Monitoring'.
22. How to reduce "Alert Fatigue"?
23. What is 'Synthetic Monitoring'?
24. How to monitor a serverless application?

12. SRE (Site Reliability Engineering) Principles

1. What is SRE? How is it different from DevOps?
2. Explain SLI, SLO, and SLA.
3. What is an 'Error Budget'?
4. What happens when the Error Budget is exhausted?
5. What is 'Toil' in SRE? How to reduce it?
6. Explain the 'On-call' lifecycle.
7. What is a 'Post-mortem'? Why should it be 'Blameless'?
8. How do you measure 'Availability'? (99.9% vs 99.99%).
9. Explain 'Capacity Planning'.
10. What is 'Chaos Engineering'? (Chaos Monkey).
11. Explain 'MTTR' and 'MTBF'.
12. How to implement 'Automated Remediation'?
13. What is 'Load Shedding'?
14. Explain 'Circuit Breaker' pattern.
15. How do you handle 'Cascading Failures'?

SRE Real-World Scenarios

17. A production service is down. Walk me through your first 5 minutes.
18. How do you define a 'Critical' incident?
19. Explain how to conduct a blameless post-mortem.
20. How do you decide the SLO for a new service?
21. Your system is 99.9% available, but users are complaining. Why?

13. Python for DevOps & Automation

1. Why is Python preferred for DevOps?
2. Explain 'Boto3' library.
3. How to list all S3 buckets using Python?
4. What is a 'Virtual Environment' in Python?
5. Explain 'Pip' and 'requirements.txt'.
6. How to parse a JSON file in Python?
7. What is the difference between a List and a Tuple?
8. How to handle exceptions in Python? (try-except).
9. Script to check if a website is up.
10. How to run a shell command from Python? (subprocess).
11. Explain 'Decorators' in Python.
12. How to read/write to a CSV file?
13. Script to find and delete old files.
14. What is 'Pytest'?
15. How to automate AWS Lambda with Python?

14. Cloud Migration & Modernization

1. Explain the 6 R's of Migration (Rehost, Replatform, Refactor, etc.).
2. What is a 'Landing Zone'?
3. How do you migrate a 1TB MySQL database to AWS RDS?
4. Explain 'AWS Migration Hub'.
5. What is 'Application Discovery Service'?
6. Challenges faced during cloud migration?
7. How to ensure security during migration?
8. What is 'Cloud Native'?
9. How to migrate On-Premise VMs to EC2? (AWS Application Migration Service).
10. Cost-benefit analysis of Cloud vs On-Premise.

15. DevSecOps & Security Compliance

1. What is 'Shift Left' security?
2. Explain 'SAST' vs 'DAST'.
3. What is 'SCA' (Software Composition Analysis)?
4. How to secure the CI/CD pipeline?
5. What is 'Infrastructure Security'?
6. Explain 'Secrets Management'.
7. What is 'IAM Best Practices'?
8. How to handle a security breach?
9. Explain 'Vulnerability Scanning'.
10. What is 'Compliance as Code'?
11. Explain 'SOC2' and 'HIPAA' compliance in cloud.
12. How to implement 'Zero Trust' architecture?
13. What is 'Penetration Testing'?
14. How to use 'SonarQube' for security?
15. Explain 'Container Security' (Runtime protection).

16. Real-Time Scenarios & Problem Solving

1. **Scenario:** A developer accidentally pushed an AWS Secret Access Key to a public GitHub repo. What are the immediate steps?
2. **Scenario:** Production is slow. You see high 'Wait I/O' on the database server. How do you resolve?
3. **Scenario:** A Jenkins build is taking 1 hour, but it used to take 10 minutes. How to optimize?
4. **Scenario:** Kubernetes cluster is running out of IP addresses for Pods. How to fix?
5. **Scenario:** You need to deploy a change to 10 microservices simultaneously. How to manage dependencies?
6. **Scenario:** Your Terraform apply failed halfway, leaving infrastructure in an inconsistent state. How to recover?
7. **Scenario:** A customer reports that the website is down only in the Europe region. Where do you start looking?
8. **Scenario:** Your company wants to reduce AWS costs by 30% in one month. What is your strategy?
9. **Scenario:** The Lead Developer refuses to write Unit Tests. How do you convince them from a DevOps perspective?
10. **Scenario:** You are asked to implement a disaster recovery plan with RPO of 1 hour. How to achieve this?

Advanced Real-Time Scenarios

11. **Scenario:** How do you migrate a legacy monolithic application to Kubernetes?
12. **Scenario:** Your Docker image build is failing due to 'No space left on device' on the CI runner.
13. **Scenario:** An Ansible playbook worked in Dev but failed in Prod due to different OS versions. How to prevent this?
14. **Scenario:** You need to implement a blue-green deployment for a stateful application. What are the risks?
15. **Scenario:** The API Gateway is rejecting requests with '429 Too Many Requests'. How to handle this as a DevOps Engineer?

17. Final Preparation & Personal Branding

- **Resume Tip:** Focus on "Impact" (e.g., "Reduced deployment time by 40%") rather than just listing tools.
- **LinkedIn:** Keep your profile active. Share your learning journey.
- **Projects:** Build an End-to-End project (Git -> Jenkins -> Docker -> K8s) and host it on GitHub.
- **Soft Skills:** Communication is as important as technical skills in DevOps.

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